

Individual Nonparametric Load Estimation Model for Power Distribution Network Planning

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Abstract—In this paper, we tackle client load estimation in a smart grid network. For that purpose, we propose an individual model based on nonparametric estimators. The model is designed for power estimations for use in network planning. Completely data-driven, the proposed methodology can be applied to both thermosensitive and nonthermosensitive clients. Real measurements collected in French distribution systems are used to validate our methodology. The proposed approach produces more reliable estimation results than the current model (termed BAGHEERA in this paper) of the French electricity company EDF does on the same data.

Index Terms—Distribution network planning, estimation model, individual load, kernel estimator, nonparametric method, smart grid, statistical technique.

NOMENCLATURE:

$P_{h,T}$	Estimated mean power of “h” hour at temperature “T” in the BAGHEERA model.
a_h, b_h, σ_h	Common coefficients of the BAGHEERA model, shared with clients in the same class.
E_0, s, E_n	Specific coefficients of the BAGHEERA model respectively stand for daily energy use, gradient, and annual energy use adjusted to the normal climatic condition.
T_{Nh}	Nonheating temperature.
T	Temperature variable.
ν_h	Estimated margin of “h” hour in the BAGHEERA model.
$P_{10\%h}$	10% upper-bound power of “h” hour estimated by the BAGHEERA model.
$c_{10\%}$	BAGHEERA 10% risk coefficient.
T'_i	Normal climatic temperature.
p	Power variable.

P_i	Power samples.
$\hat{g}_h(p)$	Kernel density estimator of the power variable distribution with smoothing parameter.
$K(\cdot)$	Normal kernel function.
S_m	CUSUM value.
$\bar{P}$	Mean value of the power samples.
T_i	Temperature samples.
h	Smoothing parameter.
N	Number of samples.
γ	Risk related to the bound limits, excess probability.
p_{max}, p_{min}	Estimated power bound limits.
$\beta_0(\cdot)$	Local constant parameter for a local polynomial regression of degree 0.
$\hat{f}_{NW}(\cdot)$	Nadaraya–Watson estimator.
$\hat{f}_{LL}(\cdot)$	Local Linear estimator.
$\alpha(\cdot), \beta(\cdot)$	Local Linear parameters for a local polynomial regression of degree 1.
$CV(\cdot)$	Cross-validation (CV) function.
n	Number of parts in the CV technique.
n_i^0	Number of elements in the i th part in the CV technique.
h_{cv}	Optimal smoothing parameter defined by CV technique.
$\hat{f}_h(\cdot), \hat{f}_{h_{cv}}(\cdot)$	A kernel-type estimator with smoothing parameter h or h_{cv} .
P_{TMB-i}	Extrapolated power samples to the TMB condition.
$\frac{\Delta p}{\Delta T}(T)$	Gradient value of the power variable with respect to the temperature variable.

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I. INTRODUCTION

DISTRIBUTION network planning involves developing a schedule of future additions that ensure the quality of energy delivery as well as the lowest possible cost. On the one

¹An average historical temperature of the same moment during past 30 years of day “i”

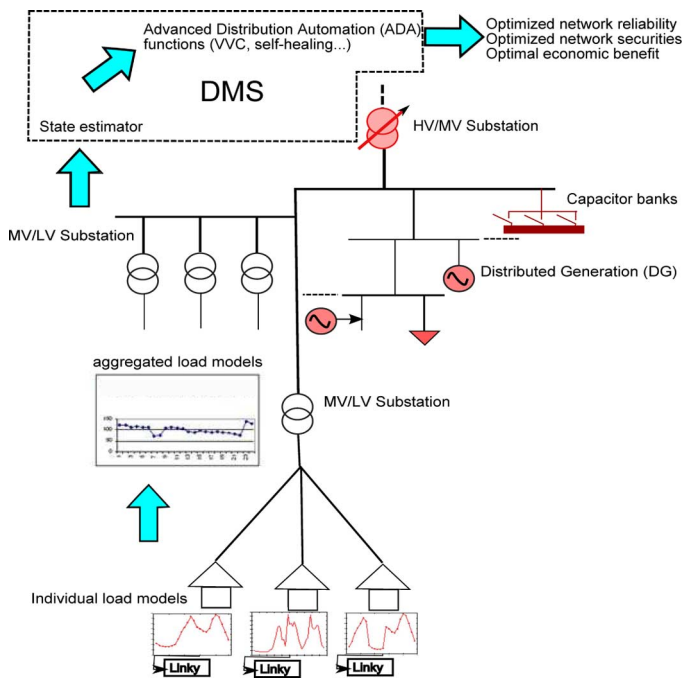


Fig. 1. Relationship between load models and DMS.

hand, the electricity infrastructure must meet the needs of peak loads. On the other hand, overdimensioned systems can be very expensive. Thus, reliable load models are required to perform distribution network calculations, such as the power flow calculations [1] in critical situations so as to identify poor electricity supply zones for investment planning.

The load models can also be applied to assess the real-time state of the network. The State Estimator (SE) [2] is the core function of any energy management system (EMS). It aims at estimating the network variables, such as the voltage magnitudes and angles. Fig. 1 shows the schematic of the relationship among load models, SE, and Advanced Distribution Automation (ADA) functions. Individual load models are aggregated to an upper voltage level. These load models as well as the network data can be considered as inputs for the SE. The network data [2] includes the information about the network topology, line resistance, reactance, tap setting, and line charging, etc. The output of the SE will lead the distribution management system (DMS) control scheduling block to perform ADA functions for operational decisions. These decisions enable the monitoring and control of various devices in the networks such as capacitor banks, distributed generations (DGs) [3], on-load tap changing transformers [4], and switches/breakers, aiming at optimizing the network reliability, securities, and economic benefit [5].

Individual load is mostly influenced by two main factors: day types and weather conditions. The daily load pattern can be very different from weekdays to weekends. Temperature is a primary weather factor that affects load variation [6], [7]. The customer model is designed for worst-case calculations [8], [9], namely, the time of maximum demand and minimum supply as well as the time of minimum demand and maximum supply contrawise [10]. As the peak loads of several customers rarely

occur at the same time, estimation of the customer consumption load at every hour is required. Uncertainty regarding load estimations also needs to be taken into account [11]. Generally, for the voltage-drop calculation, the excess probability is fixed at 10% [12]. Therefore, the objective is to define an individual customer's maximum and minimum load limits at different day types of a year with 10% excess probability.

The BAGHEERA model [13] has been developed in the late 1990s and applied ever since by the French electricity company EDF for its distribution network planning. In the paper, it is considered as a benchmark for the comparison with the proposed nonparametric model since we are capable of building the BAGHEERA models thanks to the abundant information provided by the EDF company. Moreover, data used to perform the comparison are from the French distribution network. Over 2000 demand survey load curves sampled every 10 min all over France are collected to predefine 66 classes. According to some qualitative information, such as subscribed demand and client activity category, to name a couple, every client in the French territory is associated with a predefined class. Clients in the same class share some statistically calculated coefficients defining a daily profile pattern for weekdays and for weekends. The load estimations of a specific client are computed by scaling the average shape of his particular group to his annual unit consumption. However, the smart grid paradigm introduces three ground-breaking changes to current electricity networks. First, it enables new products (electrical vehicles and renewable energy sources [14], [15]), services, and markets and thus establishes a more flexible relationship among operators, clients, and regulators [16]. Consequently, the qualitative information of each client is less and less precise. Second, customers are encouraged to modify their behavior to interact with the real-time electricity market [4], [5]. A large amount of research works regarding the smart grid concentrate on the DSM and direct load control [14], [17]–[20], which involve changing the usual consumption pattern of clients. Therefore, it is difficult to assign a client to a predefined class. Third, in 2009, the ERDF (French electricity distributor) launched the “Linky” project to install 35 million smart meters in France. These smart meters collect the power consumed by individual clients on a 30-min basis and automatically transfer the electricity consumption information to the data center [21], [22] of the ERDF at the end of every day. The data archive system is designed to keep two-year historical consumption data of every client [23]. The extensive consumption information collected by smart meters enables us to build more accurate individual estimation models. Within this context, this paper aims to design a universal individual load estimation model for the interests of distribution network planning.

Before the implementation of the smart meter, historical load data were not usually available, apart from the demand survey data of a limited number of clients. Most of the works concerning distribution network planning aimed to estimate the peak demand for a group of customers during periods of peak system demand, namely, coincident peak demand [24]. For this purpose, three categories can be identified. Some methods provide a relationship between the maximum demand of a group of clients and maximum demand of every individual

client in the group by some correction factors [10], such as coincidence factor and Velerand's formula, to name a few. Some concentrate on classification methods [12], [25], [26] to sort clients into TLPs. Others apply the end-use method [10], [27] by composing the residential customer model into appliance elementary units. The drawback of the end-use method is the extensive demand for input data. Our work, inspired by [28], proposes the building of nonparametric regressors to express the relationship between load demand and temperature and the application of nonparametric probability density function to model the random nature of load demand of a customer.

The main contributions of our work are threefold. First, the method can be adapted to any client's load regardless of his load thermosensitivity and is entirely data driven. Second, nonparametric methods are applied so that the model is independent from the client's qualitative information as well as from assumptions made on load functions. Third, the outputs of the method include both maximum and minimum daily power consumption patterns. To the best of our knowledge, building individual load models based on the smart metering data is a novel concept to the power distribution society.

The remainder of this paper is organized as follows. Section II describes the load data that are used to demonstrate both the BAGHEERA and the nonparametric models. Section III depicts the BAGHEERA model applied currently by EDF to estimate the maximum power of an individual client. Section IV details the nonparametric model procedure as well as the coordinate statistical tools. A computational example is presented and the performance of different models is discussed in Section V. Section VI concludes the paper.

II. DATA DESCRIPTION

For the sake of analysis, we use 70 load curves from different regions in France in a two-year period, from July 1, 2004, to June 30, 2006. These load curves represent the power consumption of 35 basic-option clients and 35 off-peak/on-peak option clients. Clients are numbered from 1 to 35 in each group in the database. These numbers are used simply to identify the clients in the group. The numbers have no specific meaning in terms of load consumption. Thanks to these identification numbers, each client is related to some useful information, such as geographical location, subscribed demand and activity category, to name a few. The basic option provides a uniform tariff rate for all times of the year, whereas the off-peak/on-peak option offers an eight-hours-cheaper tariff rate during a day.

As a critical factor, the temperatures of the correspondent region are provided during the same period. Figs. 2 and 3 illustrate respectively the daily average loads of clients subscribing to the two types of contract. Some data are considered as missing mainly because of temporary measurement device malfunctions or unreasonable outlier data. For example, in Fig. 2, we observe that the load curve is broken at around day 150. We can easily notice the increase in the power consumption of *client no.5* (Fig. 2) during winters due to the use of electrical heating devices. On the other hand, because alternative heating devices powered by gas, oil, or wood are used for instance, instead of electrical heating devices connected to the distribution grid, the energy consumption pattern of *client no.18* (Fig. 3) appears

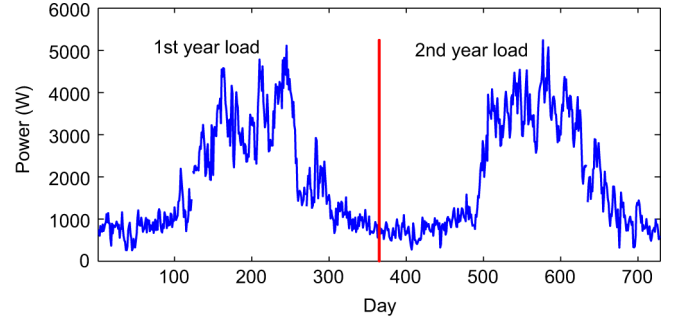


Fig. 2. Two-year daily average loads of off-peak/on-peak option of *client no.5*. The vertical line separates the two-year period.

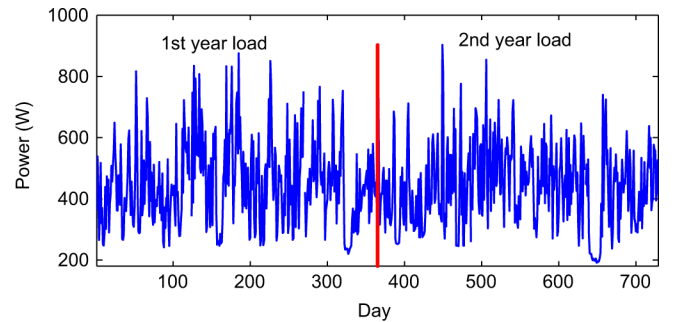


Fig. 3. Two-year daily average loads of basic option *client no.18*. The vertical line separates the two-year period.

stable during the year. Through our 70 examples, we have found several clients misplaced by the qualitative information (e.g., subscribed demand and client activity category): the observations in the data base show that the qualitative information is not a satisfactory criterion for group discrimination. As a matter of fact, relying on qualitative information to assign clients to different groups can lead to assignment errors that impact the validity of the BAGHEERA model.

III. BAGHEERA MODEL

Cooperating with electrical calculation in a worst-case situation, the BAGHEERA model estimates an individual client's load at TMB² temperature. We define TMB as the temperature threshold for every region. This latter is defined such that there exists only one day per year with an equal to or lower temperature.³ Consequently, the probability of observing a day with a temperature equal or lower than the TMB is, in general, 1/365, i.e., 0.3%. The model provides hourly estimations of weekday and weekend load profile (48 points) for every individual client. There are two components in the model: the mean power $P_{h,T}$ and the margin ν_h . For each hour "h", we have

$$P_{h,T} = a_h E_0 + b_h s(T - T_{N_h})|_{T < T_{N_h}} \quad (1)$$

$$\nu_h = \sigma_h^2 E_n. \quad (2)$$

²In French, Temperature Minimum de Base

³In practice, EDF defines the TMB based on a 30-year historical period. The probability of one day per year is an average value: in reality, during a warm year, we would probably find no daily temperature below this TMB value, while during a cold year, several daily temperatures would be found below this value.

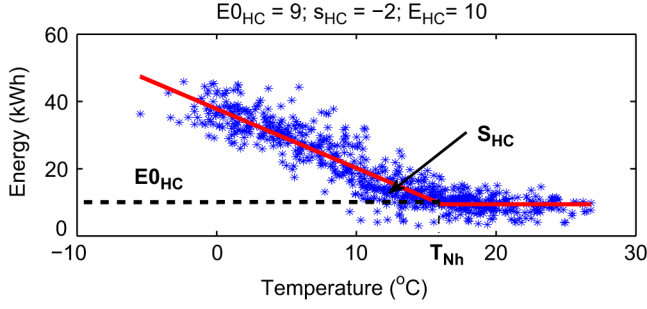


Fig. 4. Off-peak/on-peak option *client no.5*: curve fitting to off-peak daily energy use.

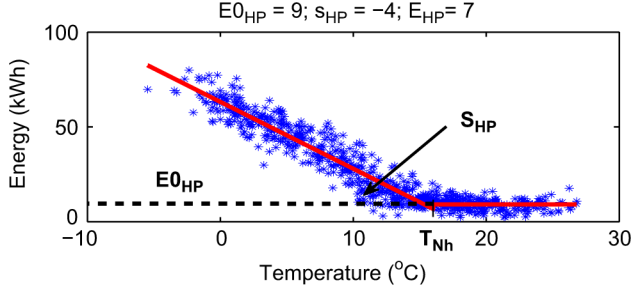


Fig. 5. Off-peak/ on-peak option *client no.5*: curve fitting to on-peak daily energy use.

All mathematical notations are defined in the Nomenclature. Equation (1) assumes that above the “non-heating temperature” (T_{Nh}), which differs from region to region, the consumed power ($a_h E_0$) is independent of the temperature and that, below this temperature, the relationship between consumed power and temperature variation is linear. The linear relationship is indicated by the gradient s .

The mean power represents the expected hourly load estimation at temperature T , while the margin represents the uncertainty of the estimation. The confidence bound fixed for the distribution network planning study is 10%, which signifies that the upper bound has a 10% chance of being exceeded. The 10% upper bound power is defined as

$$P_{10\%h} = P_{h,T} + c_{10\%}\nu_h. \quad (3)$$

Therefore, theoretically, the planning confidence bound at TMB temperature is exceeded during $8760 \text{ h/year} \times (10\% \times 0.3\%) \approx 2.5 \text{ h}$ per year.

As previously stated, the $\{E_0, s\}$ specific parameters are obtained by curve fitting the daily energy measurements with the temperature variations. An open-source MATLAB toolbox “ezyfit” [29] is applied for the curve fitting. The above T_{Nh} part data are fitted with a constant (E_0) function and the below T_{Nh} part data are fitted with a linear function ($s(T - T_{Nh}) + E$) using least square error criteria. E_n stands for the energy consumed during a normal-climatic-year period and is given by

$$E_n = 365E_0 + s \sum_{i=1}^{365} (T'_i - T_{Nh})|_{T'_i < T_{Nh}}.$$

In fact, the off-peak/on-peak option client has two electricity meters, which record separately the energy consumed during on-peak hours and that consumed during off-peak hours. Figs. 4

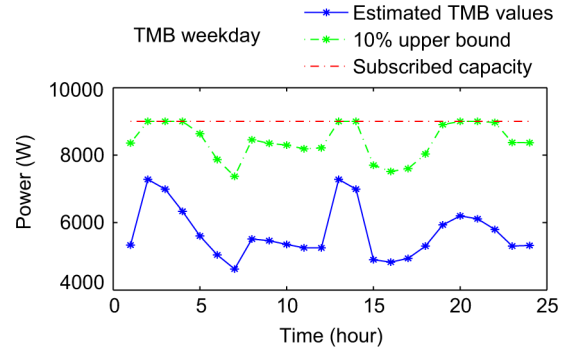


Fig. 6. Off-peak/on-peak option *client no.5*: outputs of the BAGHEERA model, TMB load estimations on weekdays.

and 5 show respectively the curve fittings to daily energy measurements of off-peak⁴ and of on-peak⁵ hours. The consumed energy is observed to vary with the temperature. In other words, the off-peak/on-peak option client often has an important gradient value. The basic option client, on the other hand, using means of heating other than electricity, stays stable with the temperature variation.

Once the parameters are calculated by curve fitting, combining with the known coefficients $\{a_h, b_h, \sigma_h\}$, (1)–(3) are carried out to calculate the 48 estimation values and their upper bounds. Notice that different groups of $\{a_h, b_h, \sigma_h\}$ values exist for weekdays and weekends, and for on-peak and off-peak hours. To calculate the off-peak/on-peak option clients’ load estimations, the on-peak estimations are replaced with the off-peak estimations during the off-peak periods. The BAGHEERA model assumes that if there is more than one off-peak period, the estimated power is the same at the beginning of each period. For example, *client no.5* has two off-peak periods: 1:00 to 7:00 and 12:00 to 14:00. We can see in Fig. 6 that two peaks at the beginning of each off-peak period have the same magnitude. The method also limits the 10% upper bound by the subscribed capacity of the client. In Fig. 6, the bound estimations higher than 9 kW are replaced by 9 kW, the subscribed capacity of *client no.5*.

IV. NONPARAMETRIC MODEL

Fig. 7 illustrates the procedure of the nonparametric model. The proposed nonparametric model is used to estimate both minimum and maximum powers (powers at TMB condition). First, the load curve of a given client is associated with the local temperature. Second, *statistical tests* are performed to examine whether the client is thermosensitive. If the client’s power consumption is not influenced by temperature, there will be no difference between minimum power and maximum (TMB) power. Thus, a *kernel density estimation* method is applied to determine the median values as well as the 10% upper boundary and the 10% lower boundary, regarding different day types. If the client is thermosensitive, the *CUMulative SUM (CUSUM) chart algorithm* is employed to separate the heating season from the non-heating season. Next, the nonheating season data are used to compute the minimum power, and the heating season data are

⁴In French, Heure Creuse (HC)

⁵In French, Heure Pointe (HP)

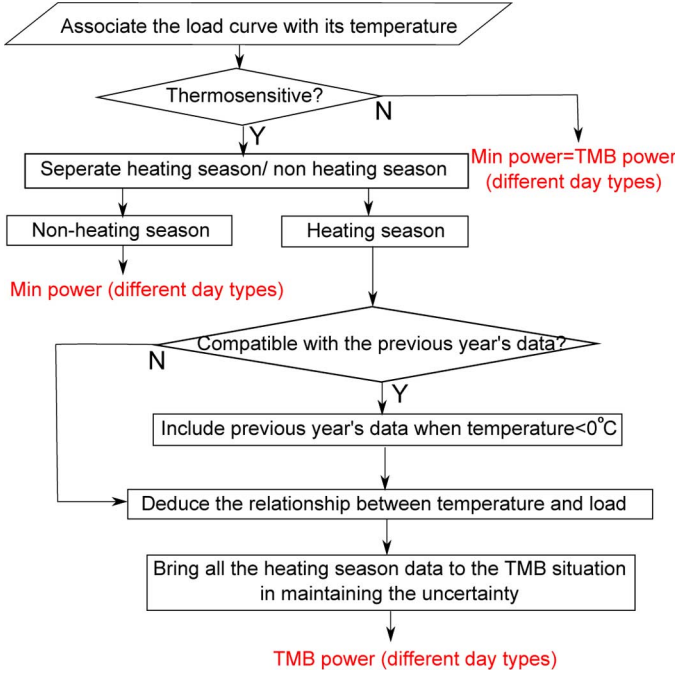


Fig. 7. Overview of the nonparametric model. The outputs of the model are the minimum and the TMB powers for different day types (weekdays, Saturdays, Sundays, and holidays).

used to compute the power at TMB temperature. Notice that the TMB temperature is the coldest temperature that happens on average one day per year, basic knowledge of electricity demand in this extreme weather condition is often insufficient. This lack of knowledge may cause a problem for the TMB load estimation. Our strategy is to include past years' consumption information if the compatibility of the data is proven by *statistical tests*. At the end of the tests, compatible data are brought in and the relationship between the temperature and the load consumption is defined with a nonparametric *kernel regression method*. The smoothing parameter required by the kernel regression method is computed by the *cross-validation technique*. Once the relationship between temperature and load consumption is established, all of the heating season data are brought to the TMB condition in maintaining their uncertainties. In the end, independently for each day type, the median power, and the excess probability boundaries can be computed at TMB temperature. These statistical tools applied in the procedure are thoroughly presented in the following subsections.

A. Statistical Tests

We aim to test if there is any significant difference between the mean of two different groups. In such a situation, *Student's t-test* for difference of means is applied. However, one assumption in carrying out *Student's t-test* is that the variance of the two populations is equal; if not, then *Welch's t-test* is used. An *F-test* can be used to test the hypothesis that the population variances are equal.

Practically speaking, at the beginning of these tests, two complementary hypotheses are set up, namely, the null hypothesis

and the alternative hypothesis. The null hypothesis often indicates that no significant difference is found between two examined groups. The decision is made by comparing a p-value indicating the probability to accept the null hypothesis with a type I error percentage referring to the risk of wrongly rejecting the null hypothesis. This type I error is often set at 0.05. In other words, if the p-value of a statistical test is greater than 0.05, one may accept the null hypothesis. Otherwise, the alternative hypothesis is accepted.

B. Kernel Density Estimation (KDE)

KDE is a nonparametric method that estimates the density directly from the data without making any parametric assumption about the underlying distribution. In our applications, KDE is used to define the γ upper and lower bounds of the data set. Let $P_i, i = 1, \dots, N$ be independent power samples drawn from a distribution $g(p)$; the kernel density estimator is defined as

$$\hat{g}_h(p) = \frac{1}{Nh} \sum_{i=1}^N K\left(\frac{p - P_i}{h}\right) \quad (4)$$

where $K(\mu) = 1/\sqrt{2\pi} \exp(-1/2\mu^2)$ is a normal kernel function.

The adaptive KDE method [30] is applied, where the smoothing parameter is chosen automatically. The reliable kernel estimator based on the smoothing properties of linear diffusion processes deals well with multimodal densities.

Thus, the power bound limits $\{p_{\min}, p_{\max}\}$ of γ excess probability, where $0 < \gamma < 1$ are defined as

$$\int_{-\infty}^{p_{\min}} \hat{g}_h(p) dp = \gamma \text{ and } \int_{-\infty}^{p_{\max}} \hat{g}_h(p) dp = 1 - \gamma. \quad (5)$$

C. Cusum Algorithm

The idea of the CUSUM algorithm is to automatically detect the breaking points in the data set that separate the heating season and the nonheating season. Afterwards, the non-heating season data are used to estimate the minimum power, and the heating season data are used to estimate the power in TMB condition.

As the name suggests, CUSUM calculates the accumulated deviations of the sampled power values $P_i, i = 1, \dots, N$ from the mean value of the power samples: $S_m = \sum_{i=1}^N (P_i - \bar{P})$.

The CUSUM plot is centered on zero. During the heating period, consumption is supposed to be superior to the mean value of the year and the CUSUM points drift upwards; during the nonheating period, consumption is likely to be inferior to the mean value of the year and the CUSUM points drift downwards. Thus, the breaking points of the two periods correspond to the maximum and the minimum values of the CUSUM plot.

D. Kernel Regression

Kernel regression is a nonparametric approach using historical data to define the relationship between load p and temperature variation T . Three kernel-type regressors are introduced here: the Nadaraya–Watson (NW) estimator, the Local Linear (LL) estimator, and the Adapted Local Linear (LL2) estimator.

Unlike the parametric estimation, the nonparametric approach imposes no assumption on the functional form and thus is much more flexible than parametric regression. The method requires the smoothing parameter h , which can be calculated by applying the cross-validation technique [31]. Estimating the conditional expectation of the random temperature variable T , the kernel regression can be defined as

$$E(p|T) = \hat{f}(T). \quad (6)$$

The estimator $\hat{f}_{NW}(T) = \beta_0(T)$ minimizes the kernel weighted least-squares

$$\sum_{i=1}^N \{P_i - \beta_0(T)\}^2 K\left(\frac{T - T_i}{h}\right). \quad (7)$$

We have the NW estimator

$$\hat{f}_{NW}(T) = \beta_0(T) = \frac{\sum_{i=1}^N K\left(\frac{T-T_i}{h}\right) P_i}{\sum_{i=1}^N K\left(\frac{T-T_i}{h}\right)}. \quad (8)$$

Equation (8) shows that the load estimation $\hat{f}_{NW}(T)$ at temperature T is a local weighted average of the historical neighborhood samples. The size of the local neighborhood and specific weights are defined by its smoothing parameter h .

However, the NW estimator applied to define the relationship between load variation and temperature has certain drawbacks. It may have a large bias in the “boundary” region due to the nonpositive sample distribution function and in the interior points due to unequal spacing of the samples [32]. Therefore, we propose another kernel-type estimator, the LL estimator of degree 1. The idea is to fit the load temperature relationship with local model $\hat{f}_{LL}(T) = \alpha(T) + \beta(T)(T - T_i)$ to minimize

$$\sum_{i=1}^N \{P_i - \beta(T)(T - T_i) - \alpha(T)\}^2 K\left(\frac{T - T_i}{h}\right). \quad (9)$$

We have the explicit expressions

$$\begin{aligned} \alpha(T) &= \frac{S_2 \sum_{i=1}^N P_i K\left(\frac{T-T_i}{h}\right) - S_1 \sum_{i=1}^N P_i K\left(\frac{T-T_i}{h}\right)(T - T_i)}{N(S_2 S_0 - S_1^2)} \\ \beta(T) &= \frac{S_0 \sum_{i=1}^N P_i K\left(\frac{T-T_i}{h}\right)(T - T_i) - S_1 \sum_{i=1}^N P_i K\left(\frac{T-T_i}{h}\right)}{N(S_2 S_0 - S_1^2)} \end{aligned} \quad (10)$$

where $S_j = \sum_{i=1}^N (T - T_i)^j K(T - T_i/h)/N$

Thus, the LL estimator can be written as

$$\hat{f}_{LL}(T) = \frac{\sum_{i=1}^N K\left(\frac{T-T_i}{h}\right)(S_2 - S_1(T - T_i))P_i}{N(S_0 S_2 - S_1^2)}. \quad (11)$$

The LL estimator preserves the linear trend and stays invariant to the first derivative to $\hat{f}_{LL}(T)$. According to the literatures [33]–[35], the LL estimator also has much better properties at the boundary than the NW estimator has. High-order estimation functions have better performance on steeper and curvier regression functions. However, when the regression function is quite flat, the NW estimator behaves better.

tion functions have better performance on steeper and curvier regression functions. However, when the regression function is quite flat, the NW estimator behaves better.

The nonparametric estimators depend mainly on samples. If there is a declining trend shown by the data, the nonparametric estimators will follow this trend. Even though in the reality, it has no actual physical meaning. Lack of samples in the cold conditions, every sample under such condition has a big influence on the nonparametric regression. Due to the imprecision in measurements or special family occasions, the few samples can sometimes cause an unreasonable trend: i.e., when the temperature decreases, the load trend decreases. However, it is known that, when the temperature drops, the electricity consumption rises because of the use of electrical heaters. Thus, we propose the LL2 estimator to correct the flaw in the trend caused by these “misleading” scarce samples. Two steps are followed to establish this LL2 estimator. First, we identify the cutoff point that separates uprising trends and declining trends of the LL estimator. Second, we filter the points on the left-hand side of the cutoff point and conserve the local trend of the points on its right-hand side.

Since the estimation is made by local averaging of historical load data in a correspondent neighborhood, it is obvious that its ability of extrapolation beyond the available historical data domain is rather limited. Hence, the estimations on the boundary are rather sensitive to the boundary samples. In our application, as TMB load needs to be estimated, but very few load samples are collected under this very cold weather condition, we propose to add previous years’ data with cold weather conditions to the estimation samples if the data characteristics are similar to those of the current year.

E. Smoothing Parameter Selection

It is well known that the result of kernel regression depends crucially on the bandwidth h selection. A large h would give an oversmoothed estimation with a large bias, whereas an h that is too small would give an undersmoothed estimation with a large variance. The cross-validation (CV) technique aims at finding the parameter h_{cv} that minimizes the CV function, defined as follows [35]:

$$CV(h) = \frac{1}{n} \sum_{i=1}^n \frac{1}{n_i^0} \sum_{j=1}^{n_i^0} (P_j - \hat{f}_h(T_j))^2. \quad (12)$$

The idea is to randomly divide the data into n equal or quasi-equal parts and to estimate the regression function by using $n-1$ parts while calculating the square error on the left part called validation data. The procedure is repeated n times until every part has been used once for the validation. The CV function is a mean square error estimator.

As mentioned in the beginning of this section, the proposed nonparametric model provides both minimum and maximum power (power at TMB condition) estimations. The minimum powers and the correspondent bound limits are directed estimated by applying the kernel density estimation on the non-heating-period data. Also relying on the kernel density estimation for the bound limit estimations, the maximum power estimations are calculated based on the heating-period data extrapolation.

⁶a line which marks the limits of available samples

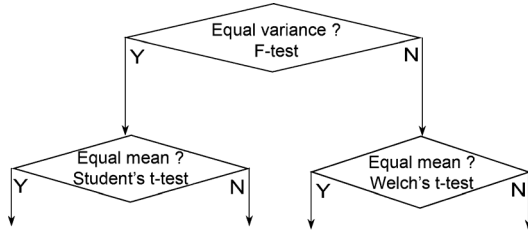


Fig. 8. Statistical tests procedure.

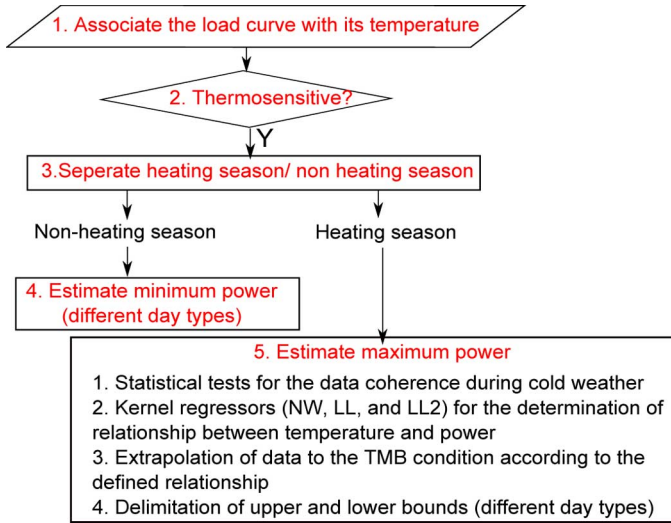


Fig. 9. Overview of the computational example steps.

olating to the TMB condition. The extrapolating relationship is defined thanks to the kernel regressors. The illustrative examples of the minimum and maximum power estimations will respectively be shown in Section V.

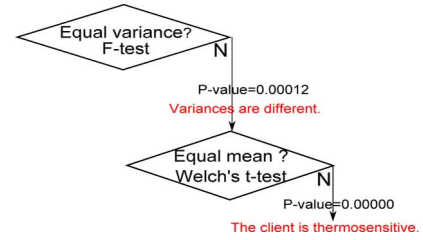
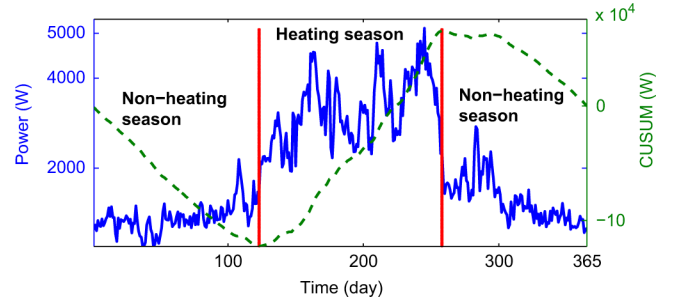
V. COMPUTATIONAL EXAMPLE

For ease of demonstration and in coherence with the previous sections, the off-peak/on-peak option client no.5 is selected as a complex example. We aim to estimate the maximum and the minimum loads of *client no.5* during the first year. Here, we first present the process in Section V-A. Section V-B illustrates and comments on the results as compared with the BAGHEERA model. All of the simulation results are run in the MATLAB environment.

A. Simulation Results

Here, we will follow a five-step process to present our computational example (Fig. 9).

- Step 1) The local temperature is associated with the client's load curve.
- Step 2) The statistical tests are carried out to check whether the examined client is thermosensitive. The statistical tests are performed by comparing the mean values between the load samples inferior to 0 °C and those superior to 20 °C. The choice of these two temperature degrees is arbitrary. As described

Fig. 10. Off-peak/on-peak option *client no.5*: statistical tests result of thermosensitive check.Fig. 11. Off-peak/on-peak option *client no.5*: separation result of one year's power data by CUSUM algorithm.

in Section II, the off-peak/on-peak option *client no.5* has probably some electrical heating device at home. Fig. 10 shows the test results. The p-value of Welch's t-test rejects the equal mean hypothesis and confirms that this client is thermosensitive.

- Step 3) The thermosensitive client's load data (one entire year) are divided chronologically into heating and nonheating seasons by the CUSUM algorithm. Fig. 11 shows the separation result. The pair of dotted lines indicates the maximum and minimum values in the CUSUM plot. They are interpreted as the breaking points of the heating and the non-heating seasons.
- Step 4) Now that the heating season and the nonheating season are separated, the minimum power load pattern can be estimated by applying the kernel density estimation on the nonheating season data. As the consumption during this period is considered independent of temperature, all days' loads are overlapped and the median value of a specific hour and its 10% excess probability upper and lower bounds are estimated by KDE [30]. The median value rather than mean value is calculated because of the former's statistical robustness to the large dispersion samples. Similar to the BAGHEERA model, three categories of different day types are made, namely, weekdays, Saturdays, Sundays, and holidays. Moreover, "Linky" in the future will collect load samples averaged every 30 min [23], and the estimation will also be made in 30-min intervals. Fig. 12 shows the result of the minimum power estimation on weekdays.

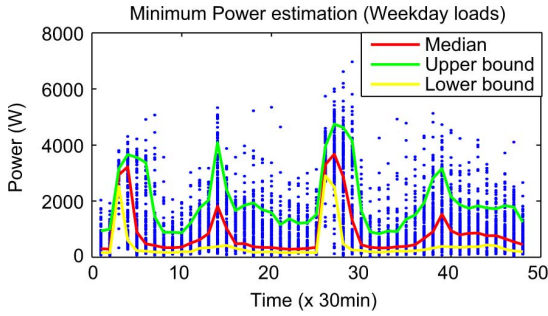


Fig. 12. Off-peak/on-peak option *client no.5*: weekday minimum power estimations. The points represent the real measurements.

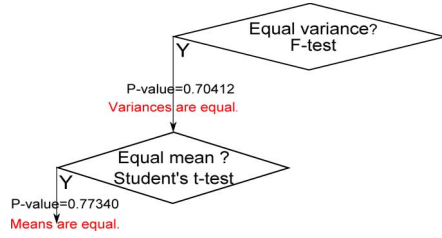


Fig. 13. Off-peak/on-peak option *client no.5*: statistical tests result for the data coherence check.

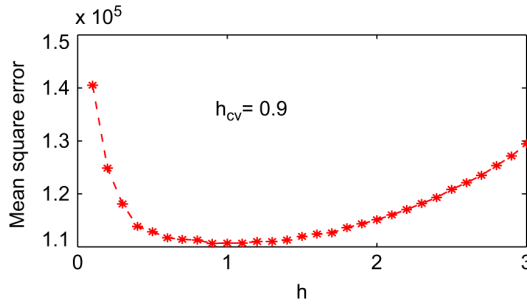


Fig. 14. Off-peak/on-peak option *client no.5*: cross-validation result on the smoothing parameter selection of the kernel estimation.

Step 5) The estimation for maximum (TMB) power of the year involves four steps.

First, statistical tests are performed between the group of load samples inferior to 0 °C of the examined year and those of the past years to check whether the consumption behavior of the client under cold weather conditions stays unchanged. In other words, whether the information collected during the extreme conditions in past years can be used for the estimation of the current year. Fig. 13 shows the statistical test results and suggests that the data of past year have similar characteristics (mean and variance values) to those of the current year. As a result, the past year's inferior 0 °C data are joined.

Second, the kernel regression method is applied to define the nonparametric relationship between power consumption and temperature. Before this is done, the smoothing parameter has to be selected. The CV function ((12)) is applied with 10 parts. Fig. 14 illustrates the Mean Square Error (MSE) variation of the different smoothing parameters. We choose $h_{cv} = 0.9$, which corresponds to the minimum MSE.

We propose herein three kernel-type estimators, referred to as: NW, LL, and LL2. These estimators are computed to deduce

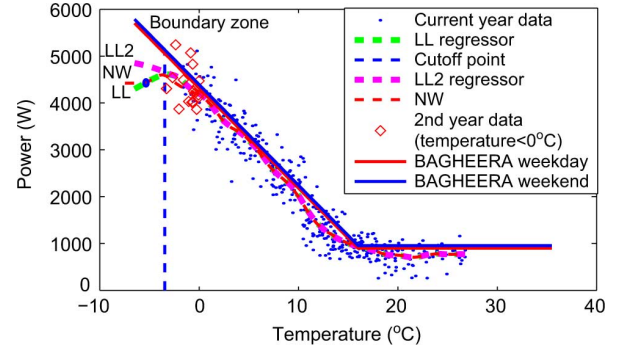


Fig. 15. Off-peak/on-peak option *client no.5*: BAGHEERA, NW, LL, and LL2 regressors, indicating the relationship between the variation of temperature and the client's daily power consumption.

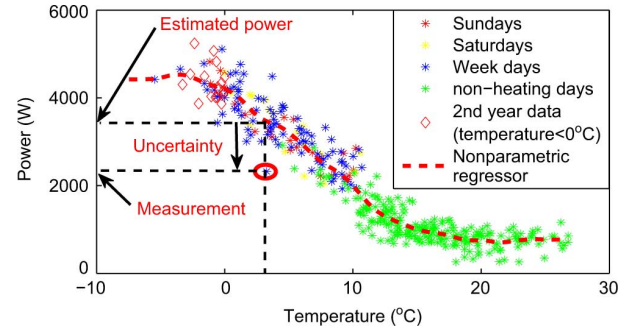


Fig. 16. Off-peak/on-peak option *client no.5*: presentation of uncertainty of the red circled sample. The dotted red curve represents the kernel estimation result of the relationship between power consumption and temperature.

the relationship shown in Fig. 15. The relationships (for both weekday and weekend) deduced by the BAGHEERA model are also shown in Fig. 15 for comparison purposes.

Notice that the NW and LL curves are rather close to each other except near the “boundary zone”. In fact, the nonparametric estimators are very sensitive to the samples on the “boundary zone.” Moreover, it is known that, when the temperature drops, the electricity consumption rises because of the use of electrical heaters. Since the lack of samples in the “boundary zone” may cause an unreasonable declining trend, the LL2 estimator is proposed to correct that flaw.

Third, all of the heating season data P_i at temperature T_i are extrapolated to the TMB condition with their uncertainty level maintained. The extrapolated power under TMB condition is defined as

$$P_{\text{TMB-}i} = \hat{f}_{h_{cv}}(\text{TMB}) - \hat{f}_{h_{cv}}(T_i) + P_i. \quad (13)$$

The uncertainty level represents the difference between the estimated power and the real measurement (Fig. 16).

Fourth, Fig. 17 shows the maximum power in the TMB condition estimated by the KDE method.

B. Comparison With the BAGHEERA Model

Here, we compare the nonparametric models with the BAGHEERA model. Since the objective is to estimate the power consumed in the TMB condition, where in our example there is no measurement, the performance criteria are hard to establish. However, the accuracy of the model can be computed

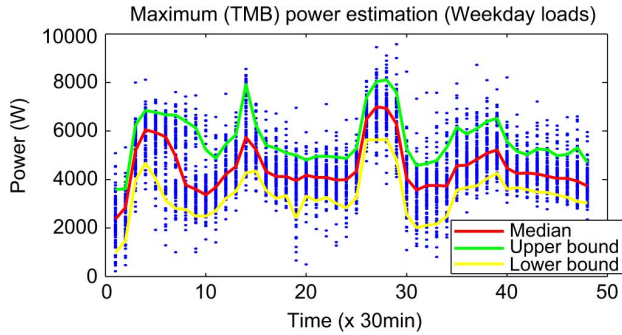


Fig. 17. Off-peak/on-peak option *client no.5*: maximum power estimation of weekday loads. The points represent the heating season data brought to the TMB condition.

in cold weather conditions instead of the TMB condition. We believe that, if the power is more accurately estimated in the neighborhood of the TMB temperature by one model, this model could possibly also provide a more credible estimation in the TMB condition.

More specifically, the first-year data are used to establish the models, respectively. The data of the second year, when the temperature is inferior to 0 °C, serve as test data. This latter correspond to 17 days, situated on the right-hand side of the cutoff line as illustrated in Fig. 15. Since test data cannot be used during the learning process, the second-year data are not joined to the first-year data to compute the kernel regression.

The sum square error (SSE) of the four models (BAGHEERA, NW, LL, and LL2) calculated on a 1-h data base on the test data are: 7.2062, 5.1648, 5.1648, and $5.1693 \times 10^8 (W^2)$.

The results indicate that the three nonparametric estimators (NW, LL, and LL2) give a better estimation than BAGHEERA does on the test data. There are two plausible explanations. One, the hypothesis on which BAGHEERA is based, that the relationship between temperature and power consumption during cold weather is linear, is not verified. Our analysis according to Fig. 16 argues that, below a certain temperature, the power consumption saturates, which is closer to the reality. Since very cold temperatures urge users to exploit their heating devices at maximum, assuming a linear relationship between the power and temperature variation does not seem very appropriate. Two, the BAGHEERA model calculates its coefficients with a group of clients. Hence, its estimation on one individual client may possibly be less accurate than a data-driven individual estimation model.

The difference in SSE between the NW and the LL estimators is fairly small with this example, as only one-year cold-temperature data were used as test data. In other words, the LL estimator presents many advantages with respect to the NW estimator. Nevertheless, the improvements with LL fit do not necessarily produce a better fitness load function in our case. We believe that, since near the TMB temperature the power function is rather flat, there is no necessity to try local polynomial regressions of higher degree.

An adapted estimator, LL2, is also proposed. It isolates the area where scarce samples are collected. As it continues the local trend drawn by the “credible data,” in our example, it

gives a higher estimation than the other two nonparametric estimators. We believe that, in the most ideal case, with sufficient data, especially with the arrival of smart meters, the three estimators converge, i.e., giving the same estimations. On the other hand, if we do not have sufficient data, particularly near the TMB region, it is better to adapt the LL2 estimator, which gives more reliable estimation values for distribution network planning. In practice, it is difficult to define the exact number of data to be “enough.” Thus, we propose a heuristic for the choice of the nonparametric estimator. The strategy is based on the gradient value $\Delta p / \Delta T|_{NW}(T)$ of the NW estimator and the gradient value of the LL estimator $\Delta p / \Delta T|_{LL}(T)$ in a small neighbourhood of TMB (around several degrees, according to the electricity distributor’s experience). The rule is as follows: if $\Delta p / \Delta T|_{NW}(T) < 0$, NW is selected; if $\Delta p / \Delta T|_{NW}(T) = 0$ and $\Delta p / \Delta T|_{LL}(T) \leq 0$, LL is selected; otherwise LL2 is selected. Since our model is individualized and completely data-driven, compared with the BAGHEERA model and most of the predefined group methodologies, the proposed methodology is more flexible in facing new types of consumption patterns. As a matter of fact, preprocessing steps such as discrimination of clients in groups are omitted, leading to a reduced complexity. Since the computational time is short (less than 45 s,⁷ the model can be frequently updated. Thus, compared with the BAGHEERA model, whose coefficients are updated every one or two years, the accuracy of the proposed estimation model is guaranteed.

VI. CONCLUSION

This paper presents a novel individual load-estimation model based on nonparametric methods for power distribution network planning. Making no hypothesis on the load function and being completely data-driven, the method is more objective in providing the accurate maximum and minimum individual load estimations. The nonparametric models are compared with the current industrial model BAGHEERA on the test data. Three different nonparametric estimators are proposed, and the heuristic for the choice of estimator is discussed. We argue that our method is more adapted and gives more reliable estimations.

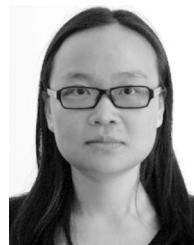
For further industrial implementation, however, more considerations need to be taken into account, such as the feasibility of 35 million individual models on French territory and the frequency of updating these models, among others.

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⁷Simulation run in MATLAB with Intel Core i5 CPU 2.67GHz, 4.00GB RAM)
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